

University of Southern Denmark
The engineering programmes in Sønderborg

Advanced Object-Oriented Programming

Practice EXAMINATION

Thursday, 7 May 2026

Problem 1 starts on the next page!

General Information

In general, for all tasks you can only use C# and the following libraries:

The .Net System library, Avalonia and CommunityToolkit.MVVM.

To avoid excessive subfolders and ensure that all your work is found and graded, only submit the requested files for each problem in a **flat folder structure**—that means a single-level folder containing all the required files. **Do not** submit a separate folder for each problem, and make sure there are no unnecessary folders like /bin/, /obj/, etc.

Your final submission folder should only contain 3 files and look **exactly** like this:

```
/AOP_Exam_Submission/  
|— Problem_2_MainWindowViewModel.cs  
|— Problem_2_MainWindow.axaml  
|— Problem_3_Comic.cs  
|— Problem_3_Program.cs
```

Remember: For Coding Task Points will be awarded regardless of the code actually “running”, as long as the code is

Problem 1 – MCQ (10 Points)

The MCQs can be found in the ItsLearning Practice Exam Plan.

Problem 2 – UI (20 Points)

In the folder `Problem_2\CircleCodeBehind\`, you will find a basic Avalonia project *CircleCodeBehind* which allows a user to change the size and color of a circle.

The Project is currently completely implemented using the CodeBehind pattern. That means, you will find the UI Layout in the `MainWindow.xaml` file and all actual functionality in the `MainWindow.xaml.cs` file.

Your task is to convert this project to the MVVM pattern so that the UI stays the same and preserves all previous functionalities.

To keep it simple just create a new MVVM Avalonia Project and copy the `MainWindow.xaml`.

Strictly follow the MVVM pattern, so only hand in the following **two files** (View and View-Model) for this problem. Any additional files will not be graded. Include only these files in your submission folder:

- `Problem_2_MainWindowViewModel.cs`
- `Problem_2_MainWindow.axaml`

Problem 3 – LINQ (20 Points)

You are given a JSON file (`Problem_3\comics.json`) that contains a database on Comics.

Add your `Program.cs` file containing the queries as `Problem_3_Program.cs` and your `Comics.cs` file containing the `Comics` class as `Problem_3_Comics.cs` to your submission folder.

3.1 Load JSON (8 Points)

3.1.1 Create a new C# Console Application and deserialise the JSON file into a list of comics. **(4 Points)**

3.1.2 Create a new class `Comic` in a `Comics.cs` file to help deserialise the contents. Make sure that all the fields are properties! `ReleaseYear` can simply be an integer. **(2 Points)**

3.1.3 Override the `ToString()` method in the `Comic` class, to print out all fields. **(2 Points)**

You will need the following imports:

```
using System.IO;
using System.Text.Json;
using System.Collections.Generic;
using System.Linq;
```

3.2 Run Queries (12 Points)

Create, run, and print the results of the following three LINQ queries on your collection. The queries should return the following:

3.2.1 All comics that were released before the year 2000. **(2 Points)**

3.2.2 The number of comics written by each author. The result should be sorted by most comics written. **(4 Points)**

3.2.3 For each year, the most active author in that year (*the author who wrote the most comics*). The result should be ordered by the year. **(6 Points)**